FASTML · ARCHITECTURE REVIEW

Ball-tree attention for jet tagging

Does a jet need every constituent to see every other one?

ErwinParTv2 | JetClass · 100k jets | 117 verification checks | 2 seeds

No ParT accuracy baseline: ParT could not be trained on this hardware. Every accuracy number here is the model against itself.

The analysis in one line

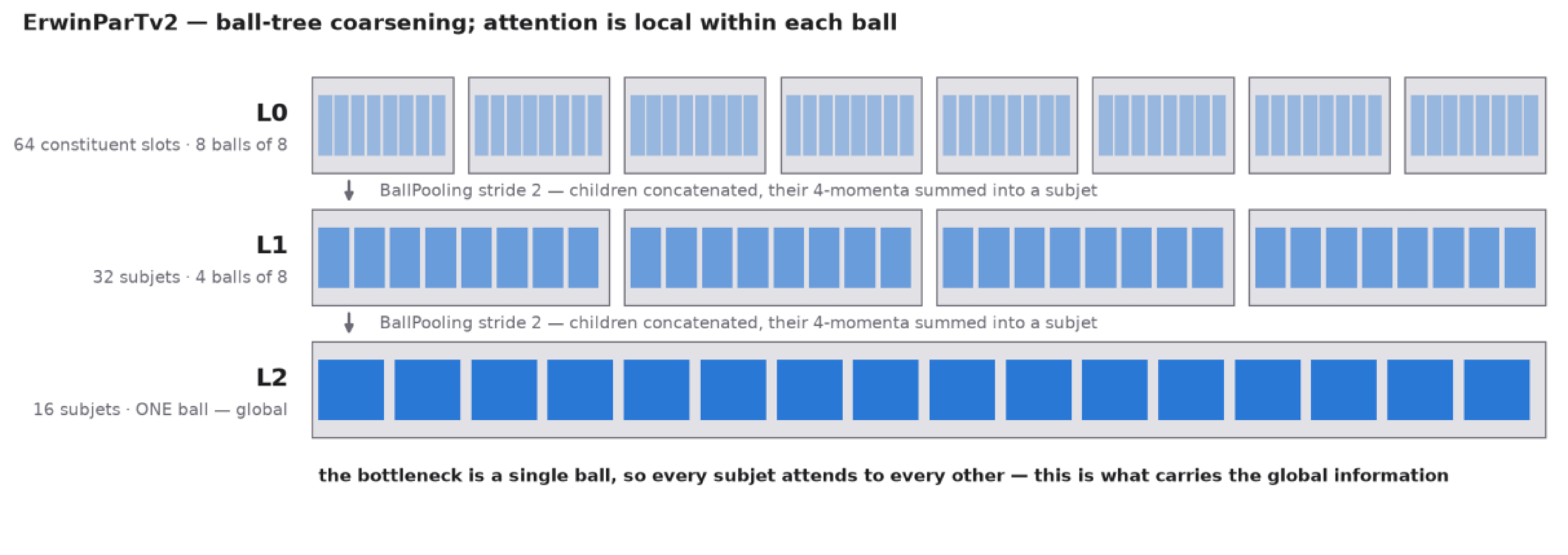
Build	Verify	Train	Ablate	Cost
ball tree + hierarchy in pure PyTorch	117 checks, S0–S9 tree parity 512/512	80k / 10k / 10k 20 epochs	ball size · U bias rotation · readout	FLOPs, memory throughput

slides 3–5 — what it is, and is the input right

slides 6–9 — does locality cost anything

slide 10 — what it buys

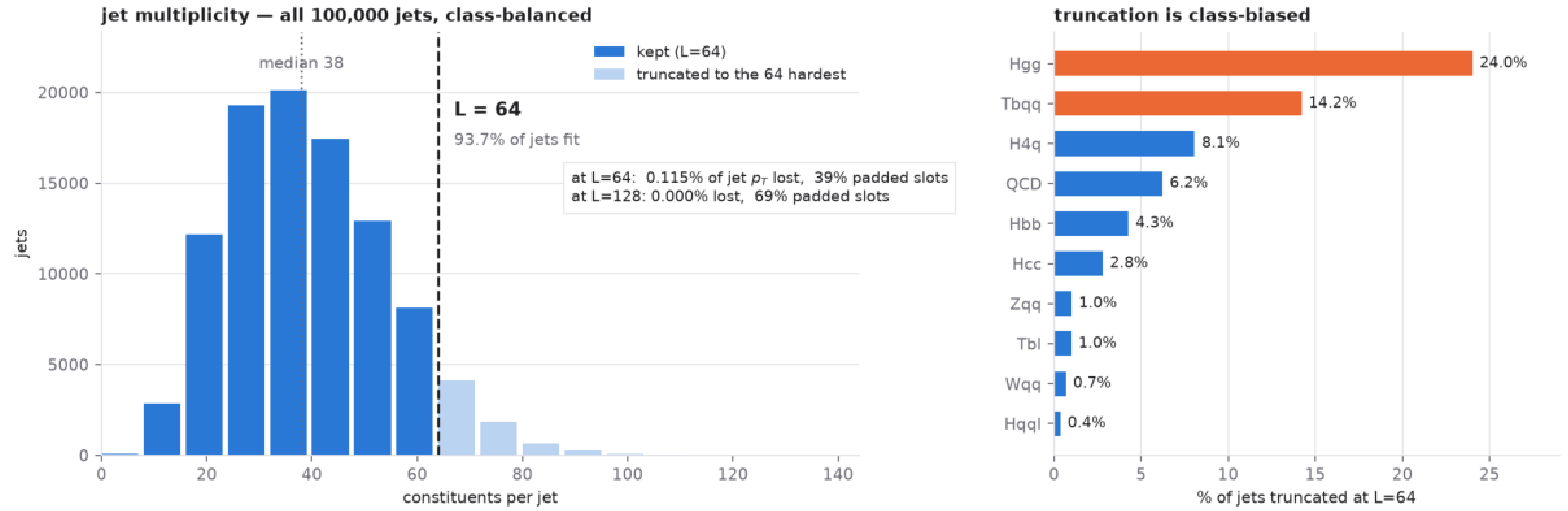
Attention is local; the hierarchy carries global information



64 > 32 > 16 nodes

bottleneck = one ball

Why 64 slots per jet



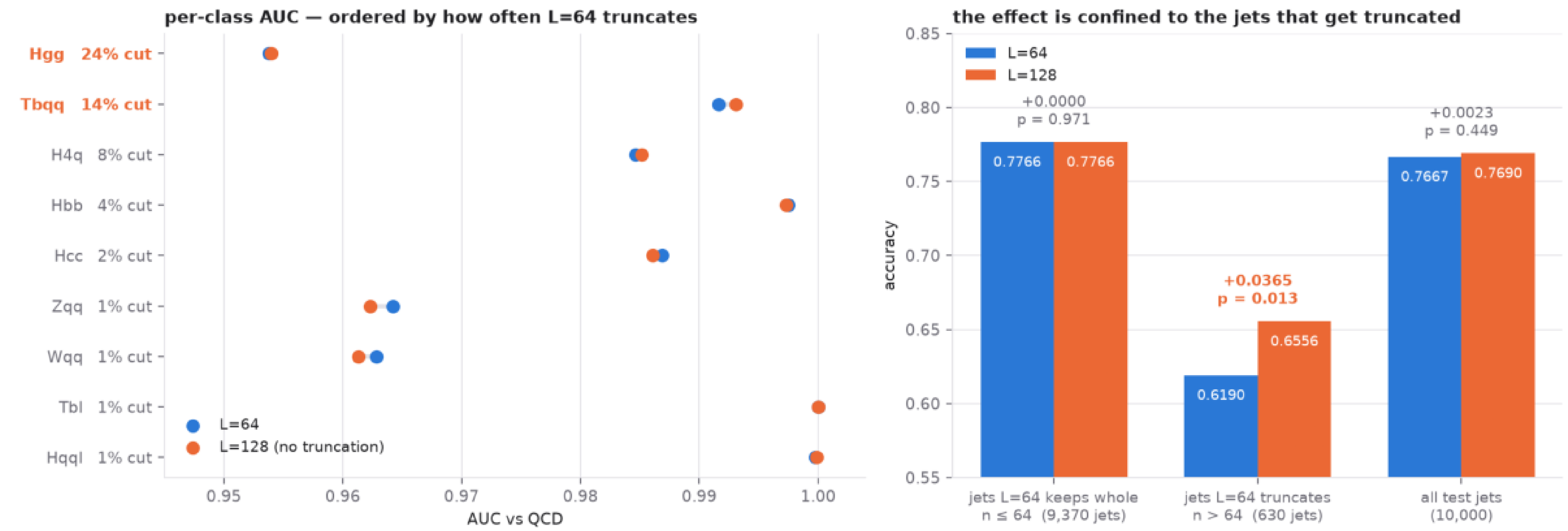
Truncation is p_T-ordered, so the discarded constituents are the softest: 1.4% of particles but only 0.115% of jet p_T. Hgg and Tbqq lose their tails most often — the L=128 control shows their AUC is unchanged regardless.

median 38

93.7% fit

Hgg loses its tail in 24% of jets

Does truncation cost us anything?



L=128 truncates nothing but costs 2.1× the training time and 2.1× the FLOPs.
On jets L=64 keeps whole the two models are IDENTICAL (p = 0.971); on the 6.3% it truncates, L=128 is 3.7 points better; over the whole test set that washes out to +0.0023 (p = 0.449).
Hgg loses its tail in 24% of jets and its AUC still moves by 0.0002 — truncation is not why Hgg is hard.

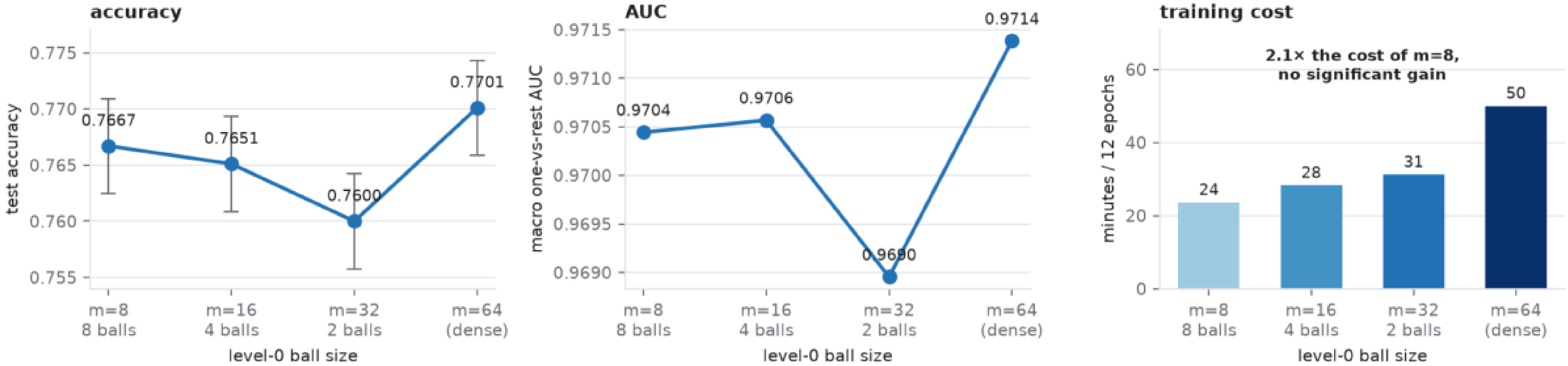
untruncated: identical, p = 0.971

truncated: +3.7 pts, p = 0.013

overall p = 0.449

Is ball locality costing us accuracy?

S11 · level-0 ball size · 80k train / 10k test · 12 epochs · 2.21M params at every m



Error bars are binomial on 10,000 test jets. Paired McNemar: m=64 vs m=8 p=0.295 (not significant); m=32 is the only arm that differs significantly from the rest (p=0.044 vs m=8, p=0.002 vs m=64).

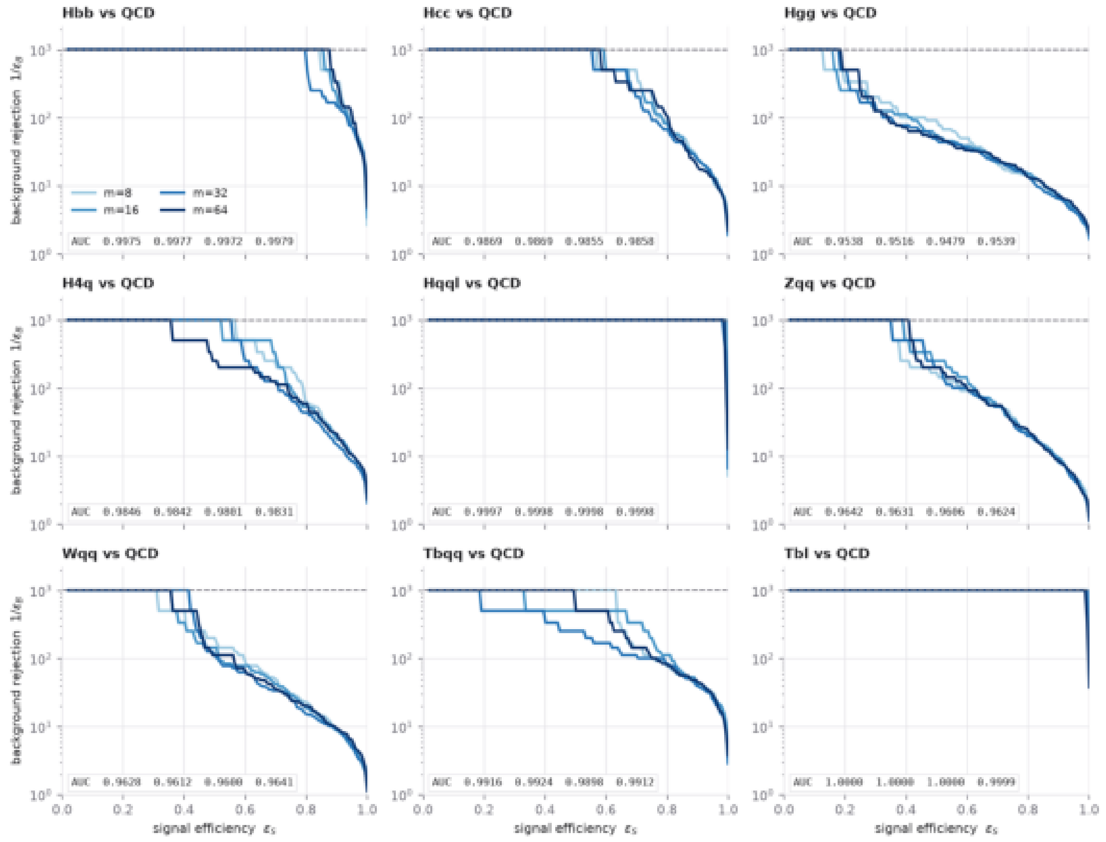
m=64 IS dense attention

vs m=8: p = 0.295, not significant

2.1x the training time

Does that hide a class-specific effect?

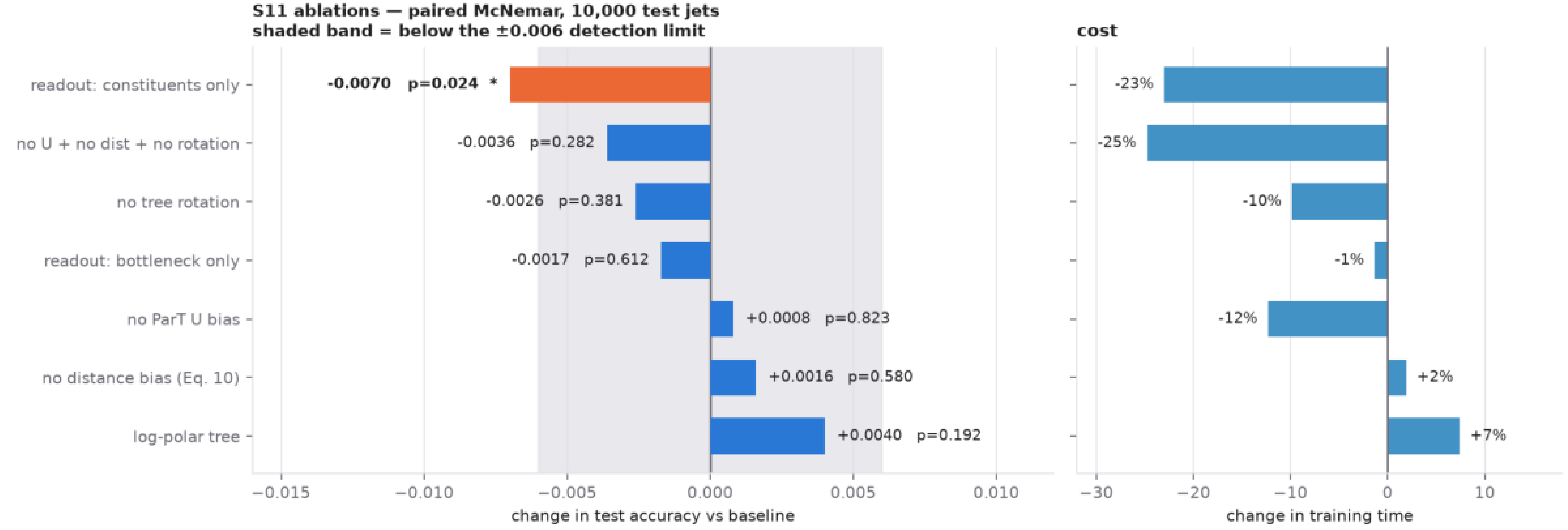
S11 · ROC by level-0 ball size · m=64 is a single ball, i.e. full dense attention



AUC values in each panel are listed m=8, 16, 32, 64. The dashed line is the ~1,000-jet measurement ceiling. Curves are largely indistinguishable: locality costs essentially nothing in tagging power.

curves overlap in all nine classes

Which parts of the architecture actually matter?



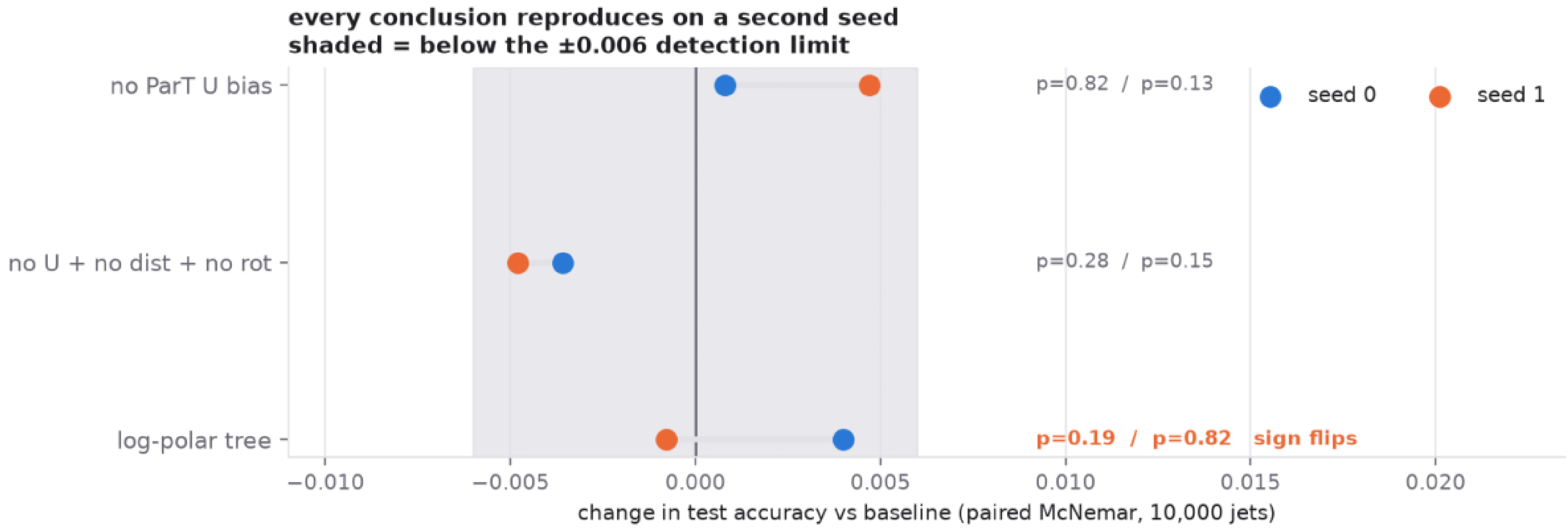
Only the constituents-only readout differs significantly from the baseline. Everything else — including ParT's interaction bias — sits inside the noise floor, individually and combined.

ParT's U bias: p = 0.823

only the readout is significant

detection limit +/- 0.006

The same conclusions on a second seed

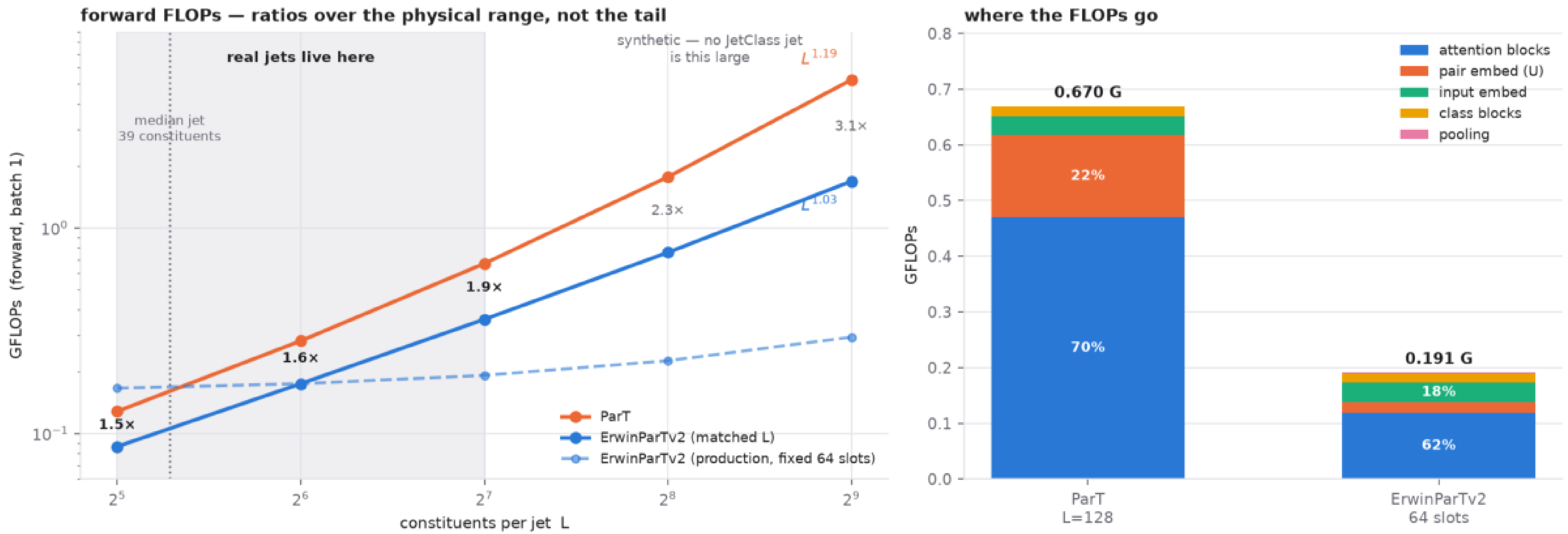


Seed changes the data split as well as initialisation, so the two seeds use different test jets — the comparison is baseline-vs-ablation WITHIN each seed.
Both architectural nulls hold twice with the same sign. The log-polar tree, the only arm that looked promising at seed 0, flips sign — it was noise.

both nulls hold twice

log-polar flips sign - it was noise

Efficiency, scoped to jets that actually exist



Bold ratios are inside the range real jets occupy; grey ones are extrapolation.

Over 32–128 the exponents are ParT $L^{1.19}$ vs ErwinParTv2 $L^{1.03}$ — near-linear against mildly superlinear, worth 1.5–1.9×, not an order of magnitude.

The dashed production curve sits ABOVE matched-L at L=32: a 32-particle jet padded to 64 slots does more work than it needs — the 36.7% cost of a fixed sequence length.

1.5-1.9x over the physical range

L^1.03 vs L^1.19

L > 128 is synthetic

What the data do and do not establish

SUPPORTED

- Dense attention is not significantly better than 8-particle balls ($p = 0.295$)
- The hierarchy is load-bearing; the readout must see coarse levels
- U bias, distance bias and rotation are all removable — confirmed on two seeds
- 1.5–1.9× fewer FLOPs; trains where ParT cannot on 16 GB
- L=64 costs 0.115% of jet pT; effect confined to 6.3% of jets

NOT SUPPORTED

- That it tags better than ParT — no baseline exists locally
- That coarse-level pair features help — predicted, measured, did not
- Effects smaller than ±0.006 accuracy — below our detection limit
- Anything at 100M-jet scale; 80k jets is 0.08% of JetClass
- That the stripped model should ship — the nulls are bounded, not zero

Where this leaves us

1 **Locality is not costing us accuracy.**
Dense attention inside our own hierarchy is no better than 8-particle balls, $p = 0.295$.

2 **The hierarchy is what matters.**
The readout must see coarse levels; everything else we inherited is removable, on two seeds.

3 **The efficiency is real but modest at jet scale.**
1.5–1.9× FLOPs — and the difference between training and not training on 16 GB.

4 **We cannot yet say it tags better.**
No local ParT baseline exists. That is the open question, not a footnote.

› **Next: the cluster run.**
kube/erwinpartv2-job.yml, full JetClass, cheap configuration first.